USC Viterbi
School of Engineering
Ming Hsieh Department of Electrical and Computer Engineering

Efficient Vision Transformer with Decorrelated Attention Head Output

Alex (Yu-Hung) Kung

Problem and Approach

- Problem: Enhancing Vision Transformer Performance
  - Goal: Improve model performance by penalizing similar output of individual attention heads
- Approach: Regularization and Decoupling
  - Method: Introduce Diversity Loss (L_{div}) term to the total loss (L_{total})
    - $L_{total} = L_{classification} + \lambda \cdot L_{div}$
    - Apply L_{div} in defreeze layers
- Base model: Fine-tune a pretrained ViT model

Diversity Loss Calculation

$$
\begin{matrix} z_1 \\ z_2 \\ z_3 \\ \vdots \\ z_n \end{matrix}
\;\Rightarrow\;
X = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ \vdots \\ z_n \end{bmatrix}
\;\Rightarrow\;
C = XX^T
\;\Rightarrow\;
\begin{matrix} S_{nxn} = \cos\_sim(X) \\ S_{ij} = \dfrac{C_{ij}}{\|z_i\|\|z_j\|} \end{matrix}
\;\Rightarrow\;
L_{div} = \|ReLU(S - I)\|_F^2
$$

Diversity Loss Formula: $L_{div} = ||ReLU(S - I)||_F^2$

$S - I$: Reduce Self Similarity

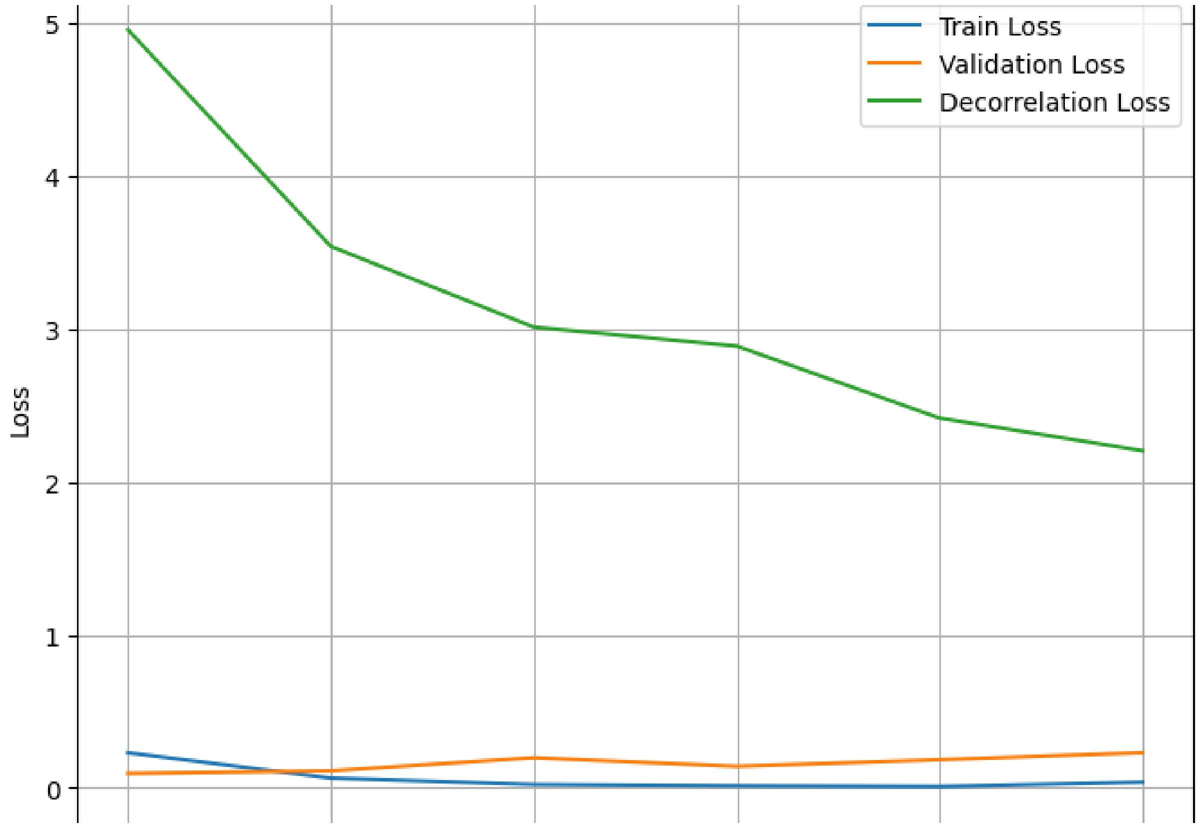
ReLU: Focus on positive similarity

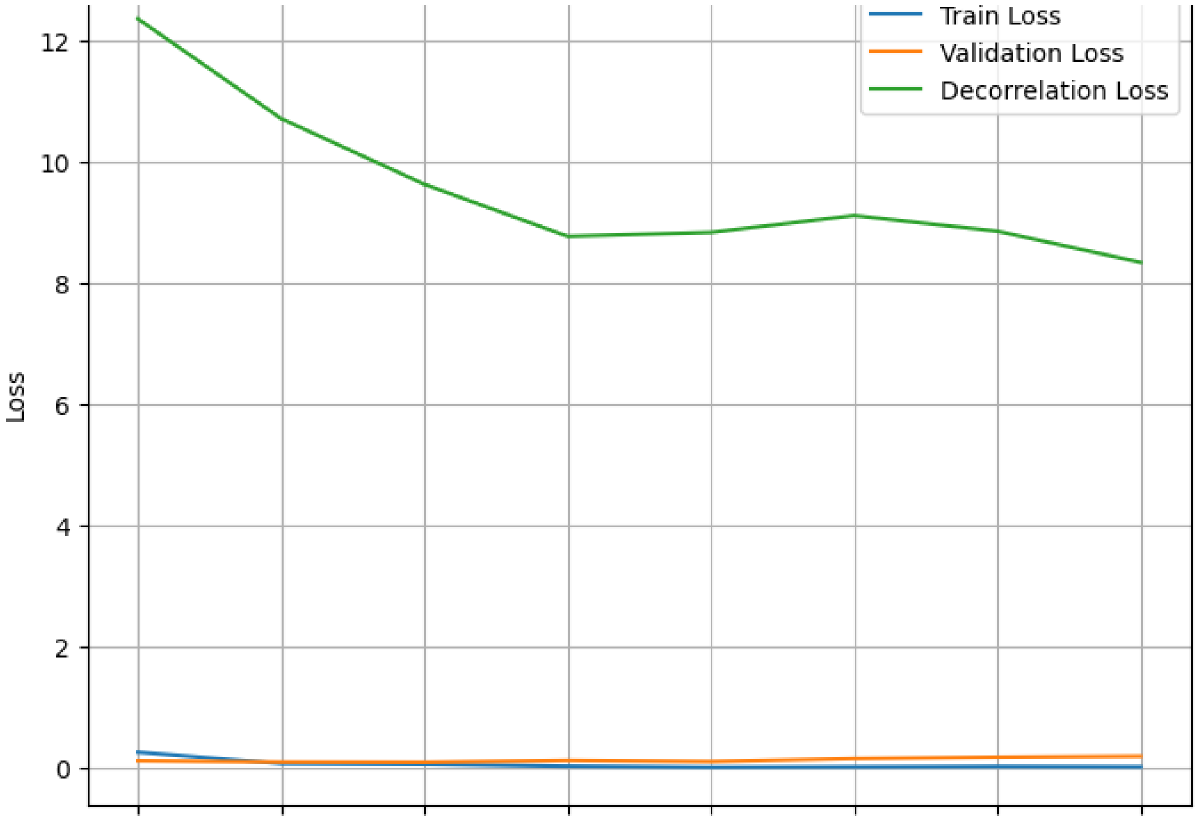
Result 1: L_{div} Limitation on CIFAR10

- Dataset: CIFAR10
- Pretrained model: ViT (google/vit-base-patch16-224-in21k)
- High-Baseline, difficult to make improvement

Model	Defreeze Layer	λ	Epoch	Train Loss	Val Loss	Accuracy
Baseline	1	0	8	0.019	0.1537	95.95%
MyModel	1	0.1	6	0.021	0.1504	96.85%
Baseline	2	0	7	0.031	0.1716	96.35%
MyModel	2	0.1	7	0.0751	0.1957	95.70%

Result 1: ViT's Natural Diversity





Decorrelation loss decreases when $\lambda = 0$

Result 2: L_{div} establish stability on CIFAR100

- Dataset: CIFAR100
- Pretrained model: ViT (google/vit-base-patch16-224-in21k)
- Train with different number of parameters, hyper-parameters, seeds, and calculate the average accuracy and its standard deviation
- Nearly 3% improvement on accuracy
- Smaller standard deviation, establish stability

Model	Defreeze layer	Lambda decorr	Learning rate	Accuracy	Standard Deviation
Baseline	4	0	1e-4	83.75%	0.06
MyModel	4	1e-4	1e-4	86.63%	0.001

Result 3: L_{div} Improvement on Waterbird

- Dataset: Waterbirds
- Pretrained model: ViT-B-32
- Trainable parameters: ~7.5M
- Train with different seeds, and calculate the average accuracy and its standard deviation

Model	Defreeze layer	Lambda decorr	Learning rate	(0, 1) accuracy	(1, 0) accuracy	Full Accuracy	Standard Deviation
Baseline	1	0	1e-4	70.17%	50.59%	82.08%	0.0149
MyModel	1	0.1	1e-4	73.61%	52.31%	83.62%	0.02

Result 3: Waterbirds - Loss vs. Ablation

- Waterbirds dataset is known for scene-base bias
- Fine-tune ViT-B-32 with L_{div}
- User-guided ablation with ViT-L-14 with CLIP

Strategy	Full Accuracy	Improvement	Analysis
L_{div} Regularization	83.62%	+1.54%	Slightly increased robustness
user-guided ablation	83.45%	+9.96%	Significantly increased robustness

Why ablation works?

- Method: Applied attention head ablation based on the TextSpan of each attention head using CLIP
- TextSpan: Extract text description of each attention head, showing what the attention head learn
- Select and ablate heads correlated with irrelevant or spurious features

Layer	Head	TextSpan	Insight
18	7	Reflective pond scene	Related to the ‘scene’
18	8	An image of Andorra	Related to ‘location / scene’

Insights and Conclusion

- **L_{div} limitation**: Statistical Diversity Loss provides limited gain for already well-performing, pretrained ViT models (CIFAR10)
- **Natural Diversity**: ViT models appear to inherently learn a degree of attention head diversity during standard training
- **Effective Strategy**: The most significant performance increase came from driven interveniton (User-Guided Ablation)
- **Conclusion**: The key to optimizing ViT performance lies in understanding and controlling the semantic roles of atttention heads, rather than solely enforcing statistical independence of their output vectors.

Summary

- L_{div} on CIFAR10 has limited impact on high-baseline ViTs
- L_{div} stablelize training accuracy in CIFAR100
- User-Guided Ablation achieved significant, semantic-driven robustness improvement with scene-based bias dataset

Thanks for listening